

Executive Summary: Statistical Testing Results

TikTok Claims Classification Project

Project Overview

TikTok's data team is developing a machine learning model to automatically classify video submissions as either claims or opinions. As part of this, project management requested an investigation to determine whether a video's creator verification status impacts its overall view count. The goal of this analysis was to implement a statistical test to check if the engagement differences between verified and unverified accounts are truly meaningful or just random chance.

Details

Key Insights

- The hypothesis test confirms with statistical p-value that is extremely smaller than the significance level 5% that there is a real difference in view counts between account types.
- Unverified accounts drive a drastically higher average view count per video compared to verified accounts

To evaluate the data, a **Two-Sample Test (Welch's t-test)** was conducted with a significance level of $\alpha = 0.05$. The data outputs revealed the following:

- **Unverified Accounts Sample Mean:** ~265,664 views
- **Verified Accounts Sample Mean:** ~91,439 views
- **Test Results:**
 - a. t-statistic = -25.50,
 - b. p-value = 2.61×10^{-120}

Because the p-value is significantly lower than 0.05, we successfully **rejected the null hypothesis**.

Next Steps

Conduct a follow-up analysis to see if unverified accounts are getting more views because they post a higher volume of "claim" videos, which our previous EDA showed naturally attract massive view spikes.